A Lean 4 Formalization of Textbook Dirichlet GRH
and an Exact Completed-L Analytic Frontier

Research Note

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Abstract

This note reports a kernel-checked Lean 4 formalization of the standard all-Dirichlet-character formulation of the Generalized Riemann Hypothesis (GRH), together with an exact chain of equivalences to zero-freeness and totality statements for analytic objects constructed from Mathlib's actual Dirichlet L-functions and completed L-functions. The development includes the reduction from imprimitive to primitive characters, completed-function zero transport, the primitive functional equation, a reflection-symmetric completed inverse pair, a centered even entire object, a principal-slit theta descent, a logarithmic derivative, and full coverage by a two-point Cayley parameterization. The final closed theorem identifies textbook Dirichlet GRH with totality of the resulting completed-L Cayley observable on the full critical parameter domain. No proof of GRH is claimed. The remaining mathematical frontier is an unconditional nonvanishing/totality theorem for that concrete analytic object; by the verified equivalences, such a theorem is GRH-strength.

Claim boundary. The Lean development proves closed equivalences and construction/coverage theorems. It does *not* prove that GRH is true. In particular, the final nonvanishing/totality assertion is left open and is formally equivalent to GRH.

Keywords. Generalized Riemann Hypothesis; Dirichlet L-functions; Lean 4; Mathlib; completed L-functions; formal verification; Cayley transform; logarithmic derivative.

1 Purpose and formal status

The purpose of the development is to remove formal ambiguity from a chain that begins with the literal textbook Dirichlet GRH and ends at one concrete analytic nonvanishing problem. The formalized layers discharge, inside Lean's kernel, the following issues:

- primitive versus imprimitive characters;
- the finite Euler factors created by change of level;
- nonvanishing of completion factors in the open critical strip;
- the completed primitive functional equation and reflected zero transport;
- reflection symmetrization and centered evenness;
- descent through the principal square root to a slit-plane theta object;
- logarithmic-derivative totality versus zero-freeness;
- exact coverage of the relevant slit domain by a two-point Cayley parameterization.

The latest verified package is

`GP_BKQR_CAYLEY_GRH_TEXTBOOK_ALL_CHARACTERS_A_TIER_LEAN_v0_10_0_FIX1.zip`.

It was checked with native Windows Lean 4.33.0. The project audit reports zero project/research axiom declarations and no `sorryAx`. Printed dependencies of the headline theorems are the standard Lean principles `propext`, `Classical.choice`, and `Quot.sound`.

The project term *A-Tier* is used in a narrow formal sense: the headline theorem is closed, with no project/research axioms, no theorem hypotheses, and no external research-parameter inputs. Universally quantified mathematical variables are part of the statement itself and are not counted as conditional inputs.

2 Textbook all-character GRH

Let χ be a Dirichlet character modulo N, and let $L(s,\chi)$ denote Mathlib's analytically continued Dirichlet L-function.

Definition 1 (Textbook Dirichlet GRH)**.** The project defines `TextbookGeneralizedRiemannHypothesis` to be the proposition

$$\forall N \geq 1,\ \forall \chi \pmod N,\ \forall s \in \mathbb{C}, \qquad L(s,\chi) = 0 \ \wedge\ 0 < \mathrm{Re}(s) < 1 \implies \mathrm{Re}(s) = \tfrac{1}{2}.$$

There is no primitivity hypothesis in this definition.

The project also contains the corresponding primitive-character formulation `GeneralizedRiemannHypothesis`. The following closed equivalence is kernel-checked.

Theorem 2 (All characters versus primitive characters)**.**

$$\boxed{\textit{TextbookGeneralizedRiemannHypothesis} \iff \textit{GeneralizedRiemannHypothesis}}$$

The Lean theorem is

TEXTBOOK_GENERALIZED_RIEMANN_HYPOTHESIS_IFF_GRH.

2.1 Imprimitive-to-primitive reduction

For an arbitrary Dirichlet character χ, Mathlib supplies an associated primitive character χ^* via `primitiveCharacter` and an exact change-of-level formula of the schematic form

$$L(s,\chi) = L(s,\chi^*) \prod_{p \mid N} (1 - \chi^*(p) p^{-s}),$$

with the finite product determined by the change in level.

The Lean development proves that every extra Euler factor is nonzero when $\mathrm{Re}(s) > 0$. The core estimate is

$$|\chi^*(p)| \leq 1, \qquad |p^{-s}| = p^{-\mathrm{Re}(s)} < 1,$$

so $\chi^*(p) p^{-s} \neq 1$. Consequently, in the open critical strip,

$$L(s,\chi) = 0 \quad \implies \quad L(s,\chi^*) = 0.$$

This is formalized as

`LFUNCTION_ZERO_IMPLIES_PRIMITIVE_LFUNCTION_ZERO_IN_CRITICAL_STRIP`.

Thus the all-character theorem is not a definitional relabeling of primitive GRH: the imprimitive reduction is explicitly proved in Lean.

3 Completed Dirichlet functions and reflection symmetry

The analytic construction uses Mathlib's actual `DirichletCharacter.LFunction` and `DirichletCharacter.completedLFunction`. In the open critical strip, the gamma/completion factor is proved nonzero, giving exact zero preservation

$$L(s,\chi) = 0 \quad \Longleftrightarrow \quad \Lambda(s,\chi) = 0.$$

For primitive χ, Mathlib's completed functional equation is re-exported and used to prove the reflected zero equivalence

$$\Lambda_\chi(1-s) = 0 \quad \Longleftrightarrow \quad \Lambda_{\chi^{-1}}(s) = 0.$$

Define the completed inverse pair

$$P_\chi(s) := \Lambda_\chi(s)\Lambda_{\chi^{-1}}(s).$$

Lean verifies an object-level functional equation for P_χ and the exact zero reflection

$$P_\chi(1-s) = 0 \quad \Longleftrightarrow \quad P_\chi(s) = 0.$$

To obtain literal reflection symmetry without choosing a square root of the root number, define

$$S_\chi(s) := P_\chi(s)P_\chi(1-s).$$

Then

$$S_\chi(1-s) = S_\chi(s),$$

and, for primitive χ, symmetrization introduces no new zeros. Centering at $1/2$ gives

$$H_\chi(w) := S_\chi\left(\tfrac{1}{2} + w\right), \qquad H_\chi(-w) = H_\chi(w).$$

This even entire object is the analytic carrier used in the subsequent descent.

4 Slit-theta descent and the critical domain

Using the principal complex square root, define

$$\Theta_\chi(z) := H_\chi(\sqrt{z}).$$

For nontrivial χ, Lean proves differentiability of Θ_χ on Mathlib's principal slit plane. Evenness removes the sign ambiguity of the square root and yields the exact square-descent identity

$$\boxed{\Theta_\chi(w^2) = H_\chi(w)}.$$

The corresponding geometry is also kernel-checked:

$$\boxed{w^2 \notin \text{slitPlane} \iff \operatorname{Re}(w) = 0.}$$

Hence the centered critical line is transformed exactly into the removed nonpositive-real slit after squaring.

Let D_{crit} denote the squared image of the centered critical strip. The development proves, per character,

$$\Theta_\chi(z) \neq 0 \quad \text{for every } z \in D_{\text{crit}} \cap \text{slitPlane}$$

if and only if the centered completed-pair RH statement holds for χ. At family level this gives

$$\boxed{\text{completed-pair slit-theta critical zero-freeness} \iff \text{GRH}.}$$

5 Logarithmic derivative and two-point Cayley observable

On the zero-free locus, the logarithmic derivative is

$$R_\chi(z) = \frac{\Theta'_\chi(z)}{\Theta_\chi(z)}.$$

The Lean formalization represents this as an `Option` value: a zero of Θ_χ is represented by non-totality rather than by an implicit division by zero. It proves

$$\boxed{\text{critical-domain totality of } R_\chi \iff \text{critical-domain zero-freeness of } \Theta_\chi.}$$

The literal two-point Cayley observable samples conjugate Cayley points $z_+(a,r)$ and $z_-(a,r)$ and combines the corresponding logarithmic derivatives. Its denominator failure is characterized exactly:

$$C_\chi(a,r) = \text{none} \iff \Theta_\chi(z_+(a,r)) = 0 \ \vee \ \Theta_\chi(z_-(a,r)) = 0.$$

A separate coverage theorem proves that the (a,r) Cayley pair covers the entire relevant critical square/slit domain. Consequently, totality of the actual two-point observable on every admissible critical sample is equivalent to zero-freeness of Θ_χ on the full critical slit domain.

6 Main closed theorem

Combining the all-character reduction with the completed-L analytic construction gives the headline equivalence.

Theorem 3 (Kernel-checked textbook-GRH/Cayley equivalence)**.** *In the verified Lean package,*

$$\boxed{\begin{array}{c}\texttt{GeneralizedCompletedPairActualCayleyObservableCriticalTotal} \\ \iff \quad \texttt{TextbookGeneralizedRiemannHypothesis}\end{array}}$$

The exact Lean theorem is

`GENERALIZED_COMPLETED_PAIR_ACTUAL_CAYLEY_OBSERVABLE_CRITICAL_TOTAL_IFF_TEXTBOOK_GRH.`

The frozen closed atlas is

`CayleyGRHStrong.TEXTBOOK_ALL_CHARACTERS_GRH_A_TIER_ATLAS`.

It simultaneously records the following closed equivalences/implication:

$$\begin{aligned}
\text{all-character textbook GRH} &\iff \text{primitive-character GRH},\\
\text{slit-theta critical zero-freeness} &\iff \text{all-character textbook GRH},\\
\text{slit-theta log-derivative critical totality} &\iff \text{all-character textbook GRH},\\
\text{actual two-point Cayley critical totality} &\iff \text{all-character textbook GRH},\\
\text{all-character textbook GRH} &\Longrightarrow \text{Mathlib's ordinary Riemann Hypothesis}.
\end{aligned}$$

7 Kernel audit and reproducibility

The native Windows Lean 4.33.0 verification of

`GP_BKQR_CAYLEY_GRH_TEXTBOOK_ALL_CHARACTERS_A_TIER_LEAN_v0_10_0_FIX1`

reported:

Audit item	Result
`PROJECT_BUILD`	`PASS`
`RUN`	`PASS`
`AUDIT_PASS`	`Main.lean`
Project/research axiom declarations	0
Research axiom count	0
`SORRYAX`	`NONE`
Printed logical dependencies	`propext`, `Classical.choice`, `Quot.sound`
Lean version	4.33.0

Package SHA-256:

`32f964156018247b884f74259e193e0242afb4caffa1f0f4d4dcad269b252a89`

The runner used native Windows Lean 4.33.0 with no network, Python, PowerShell, Lake, or WSL during verification. The all-character package is a thin build extension of the previously verified completed-pair/Cayley-coverage baseline; this is a source/build dependency, not a mathematical hypothesis of the closed theorem.

For a rapid independent audit, the most load-bearing source points are:

1. `TextbookDirichletGRH.lean`: the literal all-character definition and primitive reduction;
2. `CompletedDirichletBridge.lean`: ordinary/completed zero preservation;
3. `PrimitiveCompletedPair.lean`: functional equation and completed inverse pair;
4. `CompletedPairCayleyAnalytic.lean`: centered even object, slit theta, literal Cayley observable;

5. `CompletedPairSquareSlitGeometry.lean`: square descent and critical slit geometry;
6. `CompletedPairLogDerivativeTotality.lean`: log-derivative totality equivalence;
7. `CompletedPairCayleyCoverage.lean`: full Cayley parameter coverage;
8. `Main.lean`: headline `#check` and `#print axioms` audit.

8 The remaining research frontier

No unconditional proof of textbook GRH is contained in the package. The remaining load-bearing statement can be expressed directly as

$$\boxed{\forall N,\ \forall \chi \text{ primitive},\ \forall z \in D_{\mathrm{crit}} \cap \mathrm{slitPlane}, \qquad \Theta_\chi(z) \neq 0.}$$

By the formal equivalences above, this is equivalent to textbook Dirichlet GRH.

Equivalently, one may target unconditional totality of the logarithmic derivative Θ'_χ/Θ_χ or of the literal two-point Cayley observable on the complete admissible critical domain. At this stage, further formal work that merely introduces another equivalent predicate would add little; a genuinely new result must supply analytic information strong enough to exclude zeros/poles.

Potential directions for expert scrutiny include:

- a Pick/Herglotz or positive-kernel representation for Θ'_χ/Θ_χ;
- a de Branges-type or spectral-positivity interpretation of the centered completed pair;
- argument-principle or phase-monotonicity control on the slit domain;
- a coercive energy identity whose positivity excludes poles of the logarithmic derivative;
- a direct analytic mechanism proving totality of the literal Cayley observable.

These are proposed research directions, not results established by the Lean package.

9 Discussion

The formal contribution is therefore not a proof of GRH but an exact separation of infrastructure from substance. Definitions, existence of the analytic carriers, primitive/imprimitive reduction, completion, reflection, square/slit geometry, log-derivative semantics, and parameter coverage have all been discharged in the kernel. What remains is a single class of analytic nonvanishing statements on an explicit completed-L object.

This separation is useful even independently of the Cayley language: it provides a compact audit trail from the conventional all-character textbook formulation of GRH to one concrete analytic zero-free problem, with every reduction step machine-checked.

References

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[5] The mathlib Community, "The Lean Mathematical Library," in *CPP 2020: Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs*, 2020, pp. 367–381.

Research note prepared for expert circulation.
The Lean source archive and native verification logs should accompany the note when circulated.